

A white and black drone is flying in a forest, spraying a fine mist from four nozzles. In the foreground, three wild boars are visible: a large adult on the left, a small piglet in the center, and another adult on the right. The background is a dense forest with sunlight filtering through the trees. A small circular icon with a right-pointing arrow is in the top right corner.

Precision Wildlife Contraceptive Delivery Drone

**Overpopulation threatens endangered
species in protected areas**

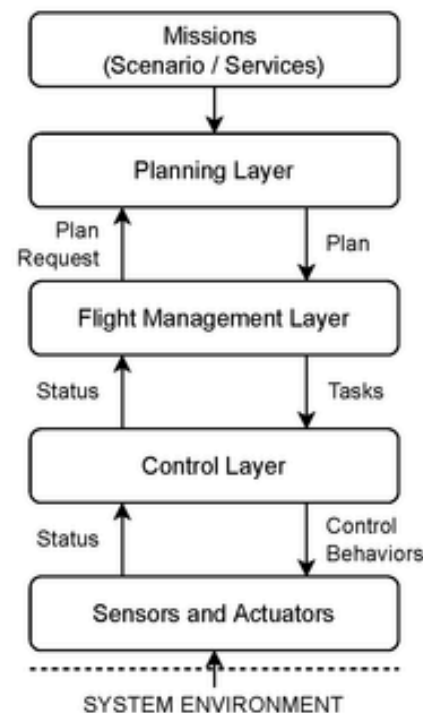
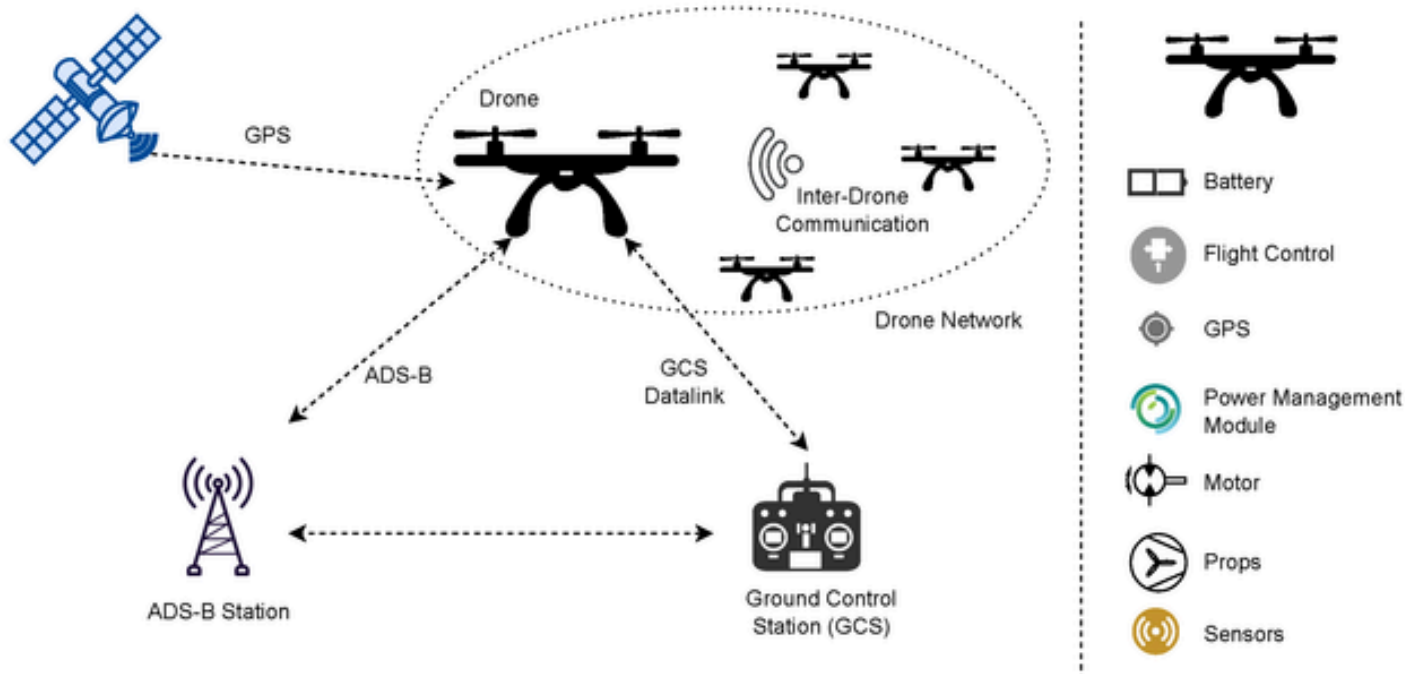
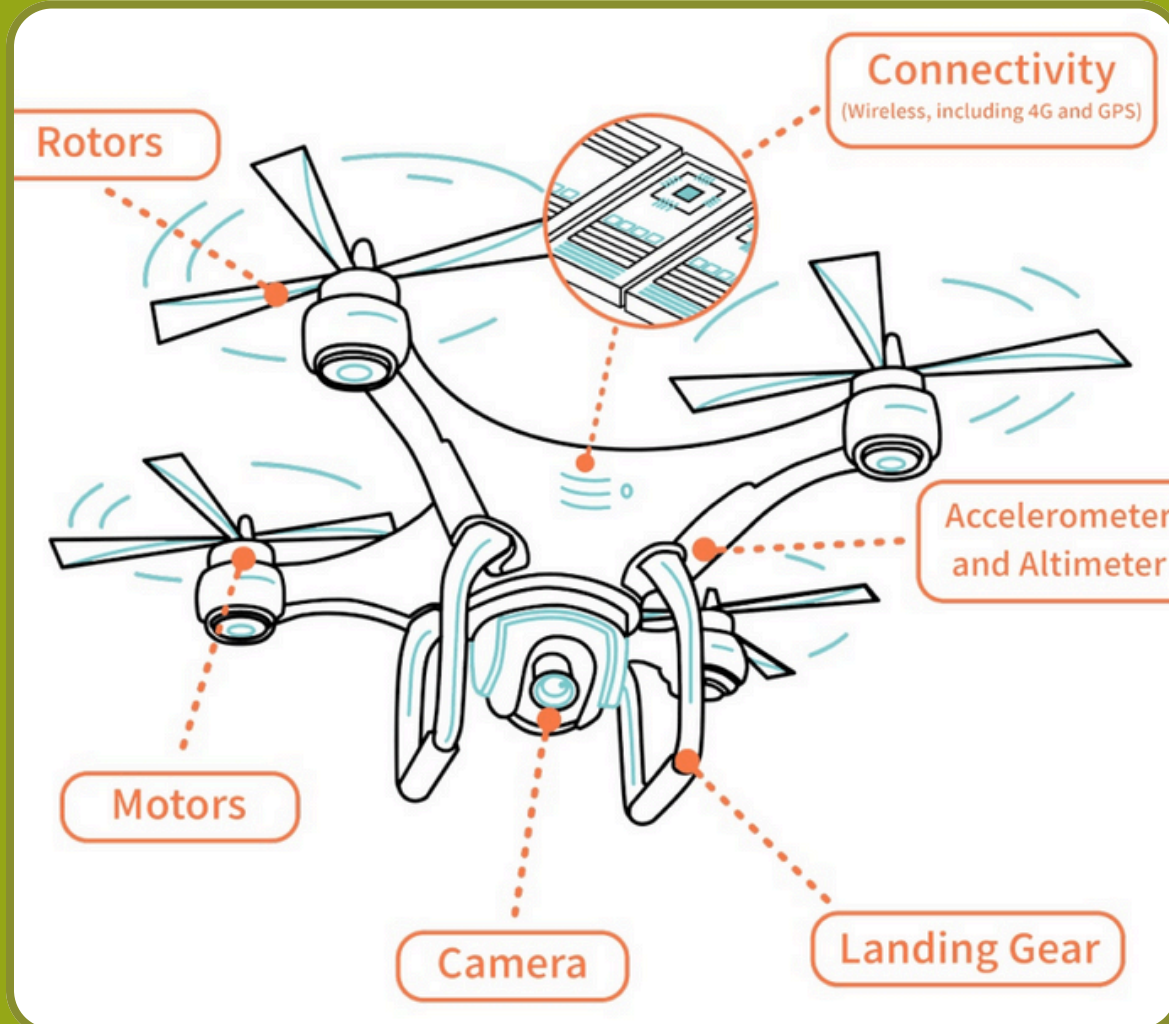
Problem Description



- Certain adaptable wildlife species experience rapid population growth within protected and buffer forest areas.
 - Overpopulation leads to habitat degradation, including destruction of forest undergrowth and soil disturbance.
 - Excessive population density causes imbalance in local ecosystems, affecting food chains and regeneration cycles.
 - Species such as wild boars (reference species) expand beyond natural limits due to lack of effective population regulation.
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- Increased wildlife density results in human–wildlife conflict near forest boundaries.
 - Degraded habitats and conflict zones indirectly threaten endangered species sharing the same ecosystem.
 - Existing population control approaches rely on manual intervention or culling, which are risky, imprecise, and disruptive.
 - There is a lack of technology-driven, precise, and non-lethal systems for selective wildlife population management.

Proposed Solution Architecture

We propose an **AI-driven autonomous drone system** designed for precision **wildlife population control**, using **wild boars** as the **reference species**. The solution focuses on **accurate animal detection**, **species** and **gender identification**, and safe, **non-invasive delivery** through **spray** or pheromone-assisted pods.



The drone operates using a structured software pipeline:

- **Thermal Camera** Feed to detect warm-bodied animals in low-light or nocturnal conditions
- **AI-Based Animal Detection** Model to **identify animal presence in real time**
- **Species Identification** to verify the target species (wild boar)
- **Gender Identification** to enable selective and **minimal intervention**
- **Pheromone-Assisted Confirmation** to improve **targeting accuracy** and **reduce non-target interaction**
- Safe Delivery Mechanism using micro-spray contact delivery or pheromone-lured bait pods, avoiding darts or physical penetration
- **GPS Logging** and Autonomous Return to **record actions** and safely exit the area

Potential for Impact

- **Stabilizes overpopulated species** without causing physical harm
- Indirectly **protects** endangered species and fragile habitats
- Reduces **habitat degradation** and **resource competition**
- Helps **minimize human-wildlife conflict** in buffer areas
- **Cost-effective** compared to repeated manual wildlife interventions
- Supports long-term, **sustainable conservation** strategies

Innovation & Concept

- Addresses wildlife overpopulation using a **non-lethal, ethical alternative to culling**
- Transforms drones from passive monitoring tools to **active conservation agents**
- Integrates **AI-driven precision targeting** instead of mass population control
- Uses **probability-based validation** for realistic and responsible decision-making
- Designed specifically for **protected zones**, ensuring controlled deployment
- **Scalable** concept adaptable to multiple species and ecosystems



Technical Feasibility

- Built using **mature AI and computer vision frameworks (TensorFlow / PyTorch)**, enabling **rapid development** and **reliable model performance**
- **Thermal–RGB data fusion algorithms** are **software-driven** and already proven in **surveillance** and wildlife **monitoring applications**
- Species classification and **gender probability estimation models** use supervised learning with confidence thresholds, making outputs measurable and defensible
- **Multi-layer software validation logic** integrates AI confidence scores, visual confirmation, and **geospatial context** to reduce false decisions
- **Decision-control software** for non-invasive delivery ensures actions are triggered only when all validation conditions are met
- **Autonomous mission, geofencing, and logging software** enables safe, traceable operation and easy performance evaluation



TECH STACK & GOOGLE TECH

System Stack (Proposed)

- Autonomous quadcopter drone
- Thermal + RGB cameras
- Onboard compute unit
- Python-based ML models for animal detection, species & gender identification
- Python control logic for decision making and safety checks
- Micro-spray / pheromone bait-pod delivery
- GPS navigation, geofencing & autonomous return
- Mission logging & monitoring

Google Technologies Used

- TensorFlow / TensorFlow Lite – ML training & edge inference
- Google Cloud Platform (GCP) – data storage & analytics
- Firebase – real-time mission monitoring dashboard
- Google Maps Platform – geofencing & route planning
- Google Stitch – rapid design of system dashboards and control interfaces

